STAFF REVIEW · H446 068 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Divide the problem into smaller subproblems, solve them, then combine their solutions to solve the original problem.

2. A state unable to reach any valid solution.

3. To stop further calls at a terminal state.

4. Popped 5; bottom-to-top [2,8].

I Do — source worked steps

1. Follow the reasoning — Return from the failed choice of4 before excluding3. The state returns to [1], then choosing4 reaches5. Recursive local totals restore on return; mutable state needs an explicit undo.

We Do — guided CHECK

1. Follow the reasoning
The alternative branch 2+7 gives a valid solution.

You Do — source mark schemes

1. 2+3+4=9 [2] overshoots [1]; return from choosing4, then exclude3 [1]; choose4 with2 to reach6 [2].

2. Check sum==target success [2], end list failure [1], choose branch recurse [1], undo/exclude branch recurse [1], combine outcomes [1].

3. Negative later values can reduce sum [2]; e.g. target4, start6 then −2 [2]; pruning at6 loses valid solution [2].

Exit Ticket — source answer

1. False. Negative later numbers could lower it.